

2

12/4/2026

* Training phase 1 (1-1M iters):

$$L = L_{\text{DINO}} + L_{\text{iROT}} + L_{\text{KOLEO}}$$

* Training phase 2 (1M - iters):

$$L = L_{\text{DINO}} + L_{\text{iROT}} + L_{\text{KOLEO}} + L_{\text{Gram}}$$

① DINO Loss Cross entropy b/w Teacher & student
* large level

$$L_{\text{DINO}} = - \sum p_t \log p_s$$

$$p_{\underline{s}}(x) = \frac{\exp[f_{\underline{s}}(x)/T_s]}{\sum_k \exp[f_{\underline{s}}(x_k)/T_s]}$$

2 - class token
 $f_{\underline{s}}$ - MLP

② iBOT loss (patch dist)

$$\mathcal{L}_{iBOT} = - \sum_i p_{t_i} \lg p_{s_i}$$

i - patch indices

* randomly mask some patches
input to student (not teacher)

* calculate loss b/w outputs
from student corresponding to
masked patches & outputs
from teacher to whom the
patches were visible.

② Kohes Regularizer

* encourages uniform span of features within a batch

* given a set of n vectors
($\underline{u}_1, \dots, \underline{u}_n$)

$$L_{\text{Kohes}} = -\frac{1}{n} \sum_{i=1}^n \log(d_{n,i})$$

$$d_{n,i} = \min_{j \neq i} \|\underline{u}_i - \underline{u}_j\|$$

* $\downarrow$ min distance b/w $\underline{u}_i$ and any other point within the batch.

④ Gram Anchors

Gram matrix $G = F \cdot F^T$

$F \in \mathbb{R}^{N \times d}$ N patches, d dim.
each entry G_{ij} = dot product of

patch i & patch j 's feature
vectors. Captures all pairwise
similarities b/w patches in
an image.

$$L_{\text{Gram}} = \|G_s - G_t\|_F^2$$

* Frobenius norm of difference
b/w student & teacher Gram
matrices.

$\|\cdot\|_F$: Frobenius norm.

$$\|A\|_F = \sqrt{\sum_i \sum_j |a_{ij}|^2}$$